# Cadence

**A Stress Monitoring & Wellness Intelligence Application for iOS**

Submitted in partial fulfilment of the requirements for the award of the degree of

**Bachelor of Technology**


**Submitted By:**

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## Bonafide Certificate

Certified that this project report titled **“Cadence: A Stress Monitoring & Wellness Intelligence Application for iOS”** is the bonafide work of **Hariom** (Enrolment No. - 226301096), who carried out the project work under my supervision during the project. To the best of my knowledge, the work reported herein does not form part of any other project report or dissertation on the basis of which a degree or award was conferred on an earlier occasion

Project Guide

Mr. Ashwini Kumar

Assistant Professor
May 27, 2026
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Signature ___________

# Declaration

I, **Hariom**, hereby solemnly declare that the project report titled **“Cadence: A Stress Monitoring & Wellness Intelligence Application for iOS”**, submitted in partial fulfilment of the requirements for the award of the degree of **Bachelor of Technology in Computer Science and Engineering**, is my original work.

I further declare that:

- This project has been carried out by me under the supervision of my project guide Mr. Ashwini Kumar.
- The work has not been submitted previously to any other university, institution, or examination body for the award of any degree, diploma, or certification.
- All sources of information used in this report have been duly acknowledged and referenced in accordance with academic ethics and plagiarism norms.
- The data presented in this report is authentic to the best of my knowledge and has not been fabricated or manipulated.
- I understand that if any part of this declaration is found to be false, my project may be rejected and disciplinary action may be taken as per institutional rules.

Place: FET,Haridwar

Date: May 27, 2026

**Student Name:** Hariom

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## Acknowledgement

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**Hariom**

## Executive Summary

This project presents **Cadence**, a native iOS wellness-intelligence application that quantifies and helps reduce daily psychological stress by fusing physiological, behavioural, and lifestyle signals into a single, transparent score. The **objective** is to overcome two persistent shortcomings of consumer wellness tools: the siloing of health data across disconnected apps, and the opacity of single-signal, heart-rate-variability-only “stress scores” that report stress only after it has occurred. The **system analysis** identified thirteen modifiable drivers of stress — spanning sleep, exercise, caffeine, screen time, diet, hydration, circadian regularity, daylight, meal timing, fasting, eating triggers, mood, and symptoms — and a requirement that every score be fully explainable. The **system design** organises these drivers into a three-tier weighted model and an MVVM-plus-service-layer architecture built on SwiftUI and SwiftData, integrating Apple HealthKit, the DeviceActivity (Screen Time) framework, ActivityKit Live Activities, and a Groq large-language-model backend for natural-language nutrition parsing. The **implementation** delivers a pure, stateless, deterministic Stress Scoring v3 engine whose every output is decomposable into its contributing drivers and modifiers, alongside food logging, fasting tracking, sleep and activity analytics, guided interventions, and home-screen widgets. **Testing** centres on pinned-scenario unit tests of the deterministic scoring core and multi-scheme build verification across the main app and three extensions. The **future scope** includes an Apple Watch companion, iCloud synchronisation, and optional on-device CoreML calibration of personal driver weights. Cadence demonstrates that a transparent, auditable, privacy-preserving wellness score can be both technically robust and clinically interpretable.

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## List of Abbreviations

API: Application Programming Interface

CoreML: Apple's on-device machine-learning framework

HPA axis: Hypothalamic-Pituitary-Adrenal axis

HRV: Heart-Rate Variability

JITAI: Just-In-Time Adaptive Intervention

LLM: Large Language Model

mHealth: Mobile Health

MVVM: Model-View-ViewModel (architectural pattern)

NLP: Natural-Language Processing

PSS: Perceived Stress Scale

RHR: Resting Heart Rate

RMSSD: Root Mean Square of Successive Differences (an HRV measure)

SDNN: Standard Deviation of Normal-to-Normal intervals (an HRV measure)

UI / UX: User Interface / User Experience

XAI: Explainable Artificial Intelligence

## Abstract

Psychological stress is a major determinant of long-term health, yet the consumer tools that purport to measure it suffer from two structural weaknesses: they silo data across disconnected applications, and they rely predominantly on a single physiological signal — heart-rate variability (HRV) — which registers the consequences of stress only after sleep has already been disrupted, by which point the window for intervention has closed. This project proposes and implements **Cadence**, a native iOS application that computes an explainable daily stress score on a 0–100 scale from thirteen behavioural, physiological, and lifestyle drivers organised into three weighted tiers. The methodology is a pure, stateless, deterministic algorithm — the **Stress Scoring v3** engine — that combines a tier-weighted driver sum with a recovery bonus, a time-ramped engagement penalty, a multi-day pattern penalty, and an HRV/resting-heart-rate physiological calibrator, while degrading gracefully when wearable data is absent. The system is built on SwiftUI and SwiftData and integrates HealthKit, the DeviceActivity framework, ActivityKit, and a Groq large-language-model backend. Evaluation through deterministic unit tests confirms that identical inputs always yield identical, fully decomposable scores, enabling the interface to surface precisely why a user's stress changed rather than merely that it changed. The results indicate that a multimodal, transparent, on-device approach offers materially greater interpretability and intervention value than opaque single-signal commercial alternatives. Cadence thus contributes a reproducible, auditable, and privacy-preserving model for consumer stress quantification.

**Keywords:** stress quantification, heart-rate variability, multimodal sensing, explainable algorithms, mobile health (mHealth), iOS, SwiftUI, on-device privacy

# Chapter 1: Introduction

## 1.1 Background and Context

Psychological stress is one of the most consequential, yet least directly observable, determinants of human health. Chronic exposure to stressors imposes a cumulative physiological cost that the biologist Bruce McEwen termed *allostatic load* — the "wear and tear" the body accumulates when its stress-response systems are activated too frequently, fail to switch off, or respond inadequately to challenge [1], [2]. Elevated allostatic load is implicated in cardiovascular disease, metabolic dysregulation, impaired immunity, and mood disorders. Because the underlying biology is gradual and silent, individuals frequently remain unaware of their accumulating stress burden until it manifests as illness or burnout.

The proliferation of smartphones and wrist-worn wearables has, in principle, created an unprecedented opportunity to observe the antecedents of stress continuously and unobtrusively. Modern devices can record heart rate, heart-rate variability, sleep architecture, physical activity, and even ambient-light exposure. In practice, however, this data remains fragmented and under-exploited. The information needed to understand a single stressful day is typically scattered across a calorie-tracking app, the operating system's health store, a screen-time dashboard, and a journaling app, leaving the user to perform the cognitively demanding task of mentally correlating them.

This project addresses that gap through **Cadence**, a native iOS application that consolidates these heterogeneous signals and synthesises them into a single, interpretable measure of daily stress. Rather than treating stress as a black-box output of a proprietary model, Cadence is built on the conviction that a wellness score is actionable only to the degree that it is *explainable*.

## 1.2 Motivation

Four specific observations motivated this work. First, **conventional wellness applications silo data.** Calories live in one application, sleep in another, mood in a third; no single surface integrates them, and the burden of synthesis falls on the user. Second,

**HRV-only stress scores are reactive rather than prospective.** Commercial recovery scores derived chiefly from overnight heart-rate variability — as popularised by devices such as Whoop, Oura, and Garmin — capture the physiological aftermath of a stressful episode, but a behavioural cause such as several hours of late-night screen exposure does not register in HRV until *after* sleep has been impaired, by which time the opportunity to intervene has passed.

Third, **commercial scores are opaque.** Their formulas are proprietary and unpublished, so a user cannot determine whether today's poor score is attributable to caffeine, insufficient sleep, or excessive evening screen time, and therefore cannot act on it with confidence. Recent scholarship on trustworthy artificial intelligence in health care argues that, for high-stakes or behaviour-shaping decisions, interpretable models are preferable to post-hoc explanations of black-box systems [20], [21]. Fourth, **existing apps describe stress rather than offering modifiable levers.** They report that stress occurred without surfacing the eating triggers, fasting windows, or circadian drift that are themselves modifiable inputs.

## 1.3 Problem Statement

The central problem this project addresses can be stated concisely: *How can a consumer mobile application compute a single, trustworthy, and explainable measure of daily psychological stress from heterogeneous physiological, behavioural, and lifestyle signals, while preserving the privacy of sensitive health data and remaining useful even when wearable physiological data is incomplete or absent?*

Embedded within this statement are several sub-problems: the fusion of signals that differ in units, reliability, and sampling frequency; the assignment of defensible relative weights to drivers of differing physiological importance; the honest communication of uncertainty when inputs are missing; and the architectural challenge of integrating multiple privileged Apple frameworks within a single coherent application.

## 1.4 Aim and Objectives

The **aim** of this project is to design, implement, and validate a transparent, multimodal, on-device stress-monitoring application for iOS. This aim is decomposed into the following objectives:

1. To design an **explainable daily stress score** (0–100) computed deterministically from a set of heterogeneous physiological, behavioural, and lifestyle inputs.
2. To present every contributing input as a **modifiable factor card**, so that each signal that affects the score is also a surface for user action.
3. To persist a **complete daily wellness record** locally on device, requiring no mandatory backend server.
4. To provide **prospective interventions** — guided breathing, progressive muscle relaxation, fasting protocols, and journaling — that allow the user to act before stress accumulates.
5. To maintain **strict medical-data privacy**, ensuring that data read from Apple HealthKit never leaves the device.
6. To ensure the algorithm **degrades gracefully**, producing a useful score even when wearable physiological data such as HRV is unavailable.

## 1.5 Scope of the Project

The scope of this project encompasses the full design and implementation of the iOS application, its deterministic scoring engine, its data model, and its integrations with HealthKit, the DeviceActivity framework, ActivityKit, and a large-language-model nutrition service. The flagship deliverable is the Stress Scoring v3 engine together with the user interface that renders its decomposition. The project explicitly does *not* claim to provide medical diagnosis; Cadence is a wellness-and-self-awareness tool, not a regulated medical device. Multi-user clinical deployment, server-side population analytics, and machine-learned weight personalisation are identified as future work rather than current scope.

## 1.6 Principal Contributions

- **A transparent, deterministic, multi-driver stress model.** In contrast to opaque single-signal commercial scores, Cadence's Stress Scoring v3 engine is a pure, stateless function whose output is fully decomposable into its thirteen drivers and four modifiers.
- **A tier-weighted fusion scheme** that encodes a physiological prior — foundational lifestyle drivers dominate, modulators refine, and subjective signals adjust — rather than naively averaging dissimilar inputs.
- **Graceful degradation with honest confidence reporting**, whereby the score is accompanied by a high/medium/low confidence level derived from input coverage, and the physiological calibrator defaults to a neutral multiplier when wearable data is absent.
- **A privacy-preserving, on-device architecture** that integrates several privileged Apple frameworks while confining sensitive health data to the device.

## 1.7 Organisation of the Report

The remainder of this report is organised as follows. **Chapter 2** surveys the scientific and technical literature underpinning stress measurement, multimodal sensing, explainable health algorithms, and existing commercial systems. **Chapter 3** presents the system analysis, formalising the requirements and detailing the design of the thirteen-driver scoring algorithm. **Chapter 4** describes the system and model design, including the software architecture, navigation model, and data model. **Chapter 5** documents the implementation, technology stack, and external integrations, with supporting figures. **Chapter 6** describes the testing strategy. **Chapter 7** presents results and a comparative discussion against commercial alternatives. **Chapter 8** concludes and outlines future scope, and **Chapter 9** lists the references.

## Chapter 2: System Study and Literature Review

This chapter reviews the body of scientific and engineering literature on which Cadence is founded. It proceeds from the physiology and measurement of stress, through the established and emerging methods of automated stress detection, to the principles of explainable health algorithms and the landscape of existing systems, concluding with the research gap that this project addresses.

### 2.1 Stress: Physiology and Measurement

Stress is most usefully understood not as a single event but as the body's adaptive response to perceived demand. The allostatic-load framework introduced by McEwen and Stellar [2] and elaborated by McEwen [1] models the cumulative cost of repeated stress activation, distinguishing several damaging patterns: too-frequent activation, failure of the response to terminate, and inadequate response that allows compensatory systems to over-activate. This framework directly motivates Cadence's multi-day pattern penalty, which penalises a sustained upward trend in stress rather than treating each day in isolation.

Because the underlying neuroendocrine activity — principally activation of the hypothalamic–pituitary–adrenal axis and the resulting cortisol secretion — is not directly observable on consumer hardware, applied systems rely on validated proxies. Subjectively, the Perceived Stress Scale of Cohen, Kamarck, and Mermelstein [3] remains the most widely used self-report instrument and provides the conceptual basis for Cadence's subjective mood and symptom drivers. Objectively, the most accessible physiological proxy on wearable hardware is heart-rate variability.

### 2.2 Heart-Rate Variability as a Stress Biomarker

Heart-rate variability (HRV) — the beat-to-beat variation in the cardiac inter-beat interval — reflects the balance between sympathetic and parasympathetic activity of the autonomic nervous system. The standards of measurement and interpretation were codified in the seminal joint report of the European Society of Cardiology and the North American Society of Pacing and Electrophysiology [4], which defined the time-domain and frequency-domain measures, including the standard deviation of normal-to-normal intervals (SDNN) used by Cadence. These standards were subsequently updated by the e-Cardiology working group and the European Heart Rhythm Association to reflect advances in signal analysis and to caution against over-interpretation of short recordings [5].

A substantial body of work links reduced HRV to elevated psychological stress. The meta-analysis of Kim et al. [6] synthesised the literature and confirmed that acute and chronic stress are reliably associated with reductions in time-domain HRV measures such as SDNN and RMSSD. This evidence justifies Cadence's use of HRV — when present from an Apple Watch — as a physiological calibrator. Crucially, however, the same literature underscores the limitation that motivates this project: HRV is a *lagging* indicator that reflects autonomic state after the fact, and short-recording HRV is noisy and confounded by posture, respiration, and measurement device [5].

## 2.3 Multimodal and Behavioural Stress Detection

The recognition that a single physiological channel is insufficient has driven a shift toward multimodal stress detection. The widely used WESAD benchmark of Schmidt et al. [7] demonstrated the value of fusing multiple sensor streams — electrocardiography, electrodermal activity, temperature, respiration, and electromyography — reporting binary stress-versus-baseline classification accuracy of approximately 93 % from chest-worn sensors across fifteen subjects. This empirical result is the strongest single justification for Cadence's thirteen-driver design over an HRV-only model.

Gjoreski et al. [8] extended multimodal detection into free-living conditions, monitoring stress with a wrist device augmented by contextual information, and established the academic precedent for fusing physiological signals with behavioural context — precisely the strategy Cadence adopts. The comprehensive review of wearable stress detection by Gedam and Paul [9], published in *IEEE Access*, catalogues the sensors, signals, and machine-learning techniques across the field and confirms that multimodal approaches consistently outperform single-signal ones, while also noting the recurring practical challenges of missing data and inter-individual variability that any deployed system must handle.

## 2.4 The Scientific Basis of the Behavioural Drivers

Cadence's departure from physiology-only models rests on a large clinical literature establishing that specific behaviours are causally implicated in stress and mood. Sleep is foundational: Leproult et al. [10] demonstrated experimentally that partial and total sleep deprivation elevate next-evening plasma cortisol by 37 % and 45 % respectively, providing direct support for assigning sleep the highest driver weight. Exercise is robustly anxiolytic; the meta-analysis of Wipfli et al. [11] across forty-nine randomised trials reports a substantial overall reduction in anxiety relative to no-treatment controls.

Caffeine, often regarded as benign, is in fact a hypothalamic–pituitary–adrenal stimulant: Lovallo et al. [12] showed that a moderate dose elevates adrenocorticotropic hormone and cortisol within an hour. Evening screen exposure carries a measurable circadian cost; Chang et al. [13] found that several nights of light-emitting e-reader use phase-delayed the melatonin onset and degraded next-morning alertness, while Gooley et al. [14] showed that ordinary room light before bedtime suppresses melatonin and shortens its duration. Diet exerts a causal influence on mood, as the landmark SMILES randomised controlled trial of Jacka et al. [15] demonstrated, and even mild dehydration impairs mood and cognition [16]. Circadian irregularity — captured by the concept of *social jetlag* introduced by Wittmann et al. [17] — is associated with worse wellbeing, time-restricted eating influences metabolic and circadian health [18], and stress itself drives reward-seeking eating through well-characterised neuroendocrine pathways [19]. Each of these findings is revisited in Chapter 3 as the justification for a specific driver.

## 2.5 Explainable and Transparent Algorithms in Health

A defining design commitment of Cadence is transparency. The survey of explainability in trustworthy health-care AI by Markus, Kors, and Rijnbeek [20] argues that intrinsically interpretable models can, in many clinical contexts, substitute for post-hoc explanations of opaque systems while better supporting clinician and patient trust. Rudin [21] makes the stronger argument that for high-stakes decisions one should prefer inherently interpretable models over black boxes altogether, since post-hoc explanations may be unfaithful to the model's actual reasoning. The medical-XAI survey of Tjoa and Guan [22], published in *IEEE Transactions on Neural Networks and Learning Systems*, surveys techniques for rendering health models interpretable. Collectively, this literature provides the principled basis for Cadence's deterministic, fully decomposable scoring engine in preference to a learned black-box scorer.

## 2.6 Mobile Health and Just-in-Time Adaptive Interventions

Cadence belongs to the field of mobile health (mHealth). A particularly relevant paradigm is the just-in-time adaptive intervention (JITAI), formalised by Nahum-Shani et al. [23], which delivers the right type and amount of support at the moment a person is both receptive and in need. Cadence's modifiable factor cards and prospective interventions operationalise this principle: by surfacing a rising stressor while it is still modifiable, the application aims to intervene within the window that HRV-only systems miss.

## 2.7 Existing Systems and Commercial Comparators

Several commercial systems compute proprietary readiness or recovery scores. Whoop derives a recovery score predominantly from overnight HRV and resting heart rate; the photoplethysmography accuracy of the device has been independently validated by Bellenger et al. [26]. Oura computes a readiness score from nocturnal physiology, and the accuracy of its heart-rate and HRV measurements has been assessed against electrocardiography by Cao et al. [27]. A six-device cross-validation by Miller et al. [28] compared the sleep and cardiac measurements of several leading wearables, including Apple Watch, Garmin, Oura, and Whoop, in healthy adults. A critical observation across this literature is that while the *measurement accuracy* of these devices has been subjected to peer review, the *scoring formulas themselves are proprietary and unpublished* — an opacity that itself constitutes a motivation for Cadence's transparent alternative.

## 2.8 Research Gap

The literature establishes three conclusions that, taken together, define the gap this project fills. First, multimodal fusion materially outperforms single-signal stress detection [7], [8], [9], yet commercial consumer scores remain predominantly HRV-centric. Second, behavioural drivers are causally and measurably implicated in stress [10]–[19], yet are largely absent from those scores. Third, transparency is increasingly recognised as essential to trustworthy health AI [20], [21], [22], yet commercial scores are opaque. Cadence addresses all three simultaneously: it is multimodal, behaviourally grounded, and fully transparent, while remaining privacy-preserving and resilient to missing data.

## Chapter 3: System Analysis

This chapter formalises the requirements derived from the problem statement of Chapter 1 and the literature of Chapter 2, and then presents the analytical core of the system: the Stress Scoring v3 algorithm. Where Chapter 4 will describe *how* the software is structured, this chapter establishes *what* it must compute and *why*.

### 3.1 Requirement Analysis

#### 3.1.1 Functional Requirements

- The system shall compute a daily stress score on a 0–100 scale, where lower values indicate lower stress, updated continuously through the day.
- The system shall decompose every score into the individual contributions of its drivers and modifiers, and present these to the user.
- The system shall read physiological and activity data from Apple HealthKit, including steps, active energy, heart rate, resting heart rate, HRV, blood pressure, respiratory rate, sleep stages, and daylight exposure.
- The system shall measure screen-time usage via the DeviceActivity framework, distinguishing total from evening usage.
- The system shall allow the user to log meals in natural language and return structured nutritional information via a large-language-model service.
- The system shall provide a fasting tracker with multiple presets and a Live Activity, and guided interventions including breathing and progressive muscle relaxation.
- The system shall persist a complete daily record locally and present long-term trends, widgets, and a configurable home layout.

#### 3.1.2 Non-Functional Requirements

- **Transparency:** every output must be auditable and reproducible; identical inputs must yield identical scores.

- **Privacy:** data read from HealthKit must never leave the device; the only permissible outbound network call is the nutrition text the user explicitly enters.
- **Resilience:** the system must produce a useful, honestly-qualified score even when wearable physiological data is missing.
- **Responsiveness:** natural-language meal parsing must complete fast enough for an interactive logging experience.
- **Maintainability:** the scoring core must be unit-testable in isolation from device frameworks.

## 3.2 Feasibility Study

**Technical feasibility** is established by the maturity of Apple's frameworks — HealthKit, DeviceActivity, ActivityKit, SwiftUI, and SwiftData — each of which provides first-party support for a required capability, and by the availability of fast hosted large-language-model inference for nutrition parsing. **Operational feasibility** is high because the application runs entirely on the user's own device with no server to provision or maintain. **Economic feasibility** is favourable: the deterministic core requires no training data, no model-hosting cost, and no recurring infrastructure beyond occasional nutrition API calls, which are cached to minimise expense.

## 3.3 The Stress Scoring v3 Algorithm

The algorithm is the flagship contribution of this project. It is implemented as a **pure, stateless, deterministic function**: it holds no internal state, produces no side effects, and returns the same result for the same inputs. View-models gather inputs and feed them to the engine, which returns an array of per-driver results that the interface renders directly. This design makes the engine fully unit-testable without any device framework, and makes every score auditable.

### 3.3.1 Driver Tiers

Thirteen drivers are partitioned across three tiers that together sum to a 100-point behavioural base score. The partition encodes a physiological prior, discussed in Section

3.3.6, that foundational lifestyle factors dominate next-day stress while subjective signals refine it. Table 3.1 lists the tiers, their aggregate weights, and their constituent drivers.

| Tier | Weight | Drivers (point weight) |
|---|---|---|
| A — Foundational | 60 pts | Sleep (20), Exercise (12), Caffeine (10), Screen Time (10), Diet (8) |
| B — Modulators | 25 pts | Hydration (5), Circadian (5), Eating Triggers (5), Meal Timing (4), Daylight (3), Fasting (3) |
| C — Subjective | 15 pts | Mood (8), Symptoms (7) |

***Table 3.1*** *— The three driver tiers and their constituent point weights (summing to 100).*

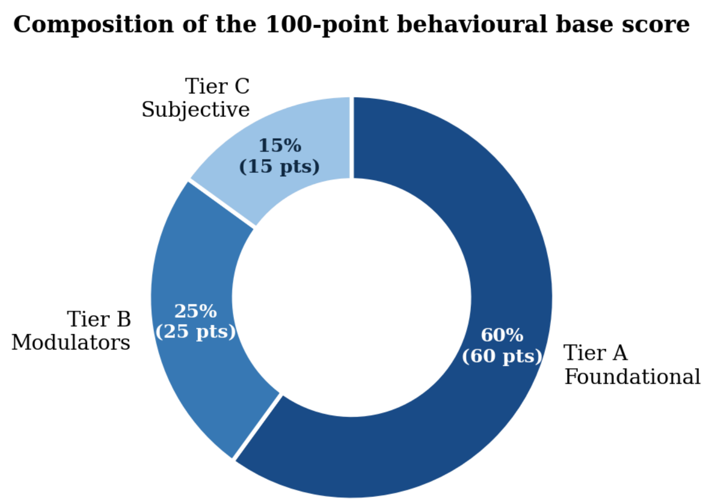


***Figure 3.1*** *— Composition of the 100-point behavioural base score by tier.*

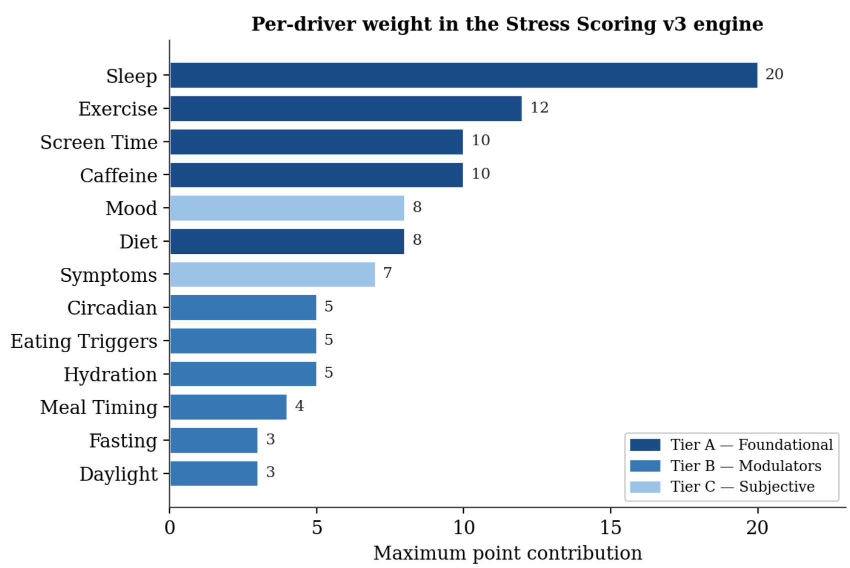


***Figure 3.2*** *— Maximum point contribution of each of the thirteen drivers, coloured by tier.*

### 3.3.2 Justification of Individual Drivers

Each driver weight is grounded in the clinical literature reviewed in Chapter 2. **Sleep** receives the largest weight (20) because experimental evidence shows sleep loss directly elevates next-evening cortisol [10]. **Exercise** (12) reflects the robust anxiolytic effect demonstrated across randomised trials [11]. **Caffeine** (10) is weighted as a genuine physiological stressor rather than a neutral stimulant, given its demonstrated stimulation of the stress axis [12]. **Screen Time** (10) is weighted heavily — with evening usage weighted more than daytime — because of the established circadian cost of evening light exposure [13], [14]. **Diet** (8) reflects the causal influence of dietary quality on mood established by the SMILES trial [15].

Among the modulators, **Hydration** (5) reflects the mood and cognitive effects of even mild dehydration [16]; **Circadian** regularity (5) penalises the social-jetlag pattern associated with poorer wellbeing [17]; **Eating Triggers** (5) capture stress-driven reward eating [19]; **Meal Timing** (4) and **Fasting** (3) reflect the metabolic and circadian relevance of when one eats [18]; and **Daylight** (3) reflects the mood benefit of light exposure. The subjective tier — **Mood** (8) and **Symptoms** (7) — operationalises validated self-report constructs [3], allowing the user's own experience to refine the objective signals.

### 3.3.3 Modifiers

After the driver sum is computed, four modifiers are applied to convert a raw behavioural total into a calibrated final score:

7. **Recovery Bonus** (capped at −10 points): subtracted for completed guided interventions — breathing, progressive muscle relaxation, journaling — and for positive, mindful mood entries, rewarding active self-regulation.
8. **Engagement Penalty** (capped at +18 points): a time-ramped penalty that is small after a single missed day of logging and escalates after several consecutive days without input, reflecting the reduced reliability of a stale record.
9. **Multi-Day Pattern Penalty** (capped at +12 points): added when a rising stress trend is detected over a rolling three-to-seven-day window, operationalising the

allostatic-load principle that sustained activation is more damaging than isolated spikes [1].

10. **HRV / RHR Calibrator** (a multiplier of at least 1.0): when Apple Watch data is available, divergence of HRV and resting heart rate from the user's personal baseline scales the behavioural score upward [6]. When no wearable data is present, the multiplier is exactly 1.0, so the algorithm degrades gracefully.

### 3.3.4 Final Score Formula

The final score combines the driver sum and modifiers as follows, with the result clamped to the valid range:

```
finalScore = clamp(
    (driverSum + recoveryBonus) × calibrator
        - engagementPenalty
        - patternPenalty,
    0, 100
)
```

The order of operations is deliberate: the recovery bonus is applied to the behavioural sum *before* the physiological calibrator, so that genuine self-regulation is amplified or attenuated in proportion to the user's measured autonomic state, while the engagement and pattern penalties are additive corrections applied afterward.

### 3.3.5 Confidence Estimation

Because missing inputs are the norm rather than the exception in real use, the engine reports a confidence level alongside every score. Coverage is computed per driver — did usable input exist for it today? — and the aggregate coverage is mapped to a qualitative level, as shown in Table 3.2. This level is surfaced verbatim in the interface; a low-confidence score is never silently presented as though it were certain.

| Aggregate driver coverage | Reported confidence |
| --- | --- |
| ≥ 70 % of drivers covered | High |
| 40 % – 70 % | Medium |
| < 40 % | Low |

***Table 3.2*** *— Mapping from input coverage to the reported confidence level.*

### 3.3.6 Why Tier-Weighting Rather Than Averaging

A naive alternative would average all available drivers. This is rejected because averaging treats dissimilar inputs as equivalent: a missed night of sleep would be weighted identically to a missed daylight log, despite the former having an order of magnitude greater impact on next-day stress [10]. Tier-weighting instead encodes a physiological prior in which foundational drivers dominate, modulators refine, and subjective signals adjust. This prior is defensible precisely because it is explicit and auditable, in keeping with the principles of interpretable health modelling [20], [21].

### 3.3.7 Determinism and Auditability

The engine contains no machine-learned component and no stochastic element; identical inputs always produce identical outputs. This determinism yields two benefits central to the project's thesis. For the **user**, it means the question "why did my score change?" always has a precise, reproducible answer expressible as a difference in driver contributions. For any future **clinical or academic review**, it means the entire model can be audited from a single table of inputs, satisfying the transparency standard advocated for high-stakes health algorithms [21], [22].

### 3.3.8 Worked Example

To make the computation concrete, consider a moderately stressful but well-logged day on which the user completed one guided breathing session, wears an Apple Watch reporting heart-rate variability slightly below their personal baseline, and is in the third day of a gently rising stress trend. Table 3.3 lists each driver's contribution to the stress total (a higher contribution indicates more stress for that driver).

| Driver | Contribution | Driver | Contribution |
|---|---|---|---|
| Sleep | 14 / 20 | Meal Timing | 2 / 4 |
| Exercise | 8 / 12 | Daylight | 1 / 3 |
| Caffeine | 7 / 10 | Fasting | 1 / 3 |
| Screen Time | 8 / 10 | Mood | 4 / 8 |
| Diet | 4 / 8 | Symptoms | 2 / 7 |
| Hydration | 2 / 5 | — | — |
| Circadian | 3 / 5 | — | — |
| Eating Triggers | 2 / 5 | Driver sum | 58 / 100 |

*Table 3.3 — Per-driver contributions for the worked example, summing to a driver total of 58.*

The four modifiers are then applied. The completed breathing session yields a **recovery bonus of −4** (within the −10 cap). The Apple Watch divergence produces a **calibrator of ×1.08**. Because the user logged today, the **engagement penalty is 0**; because a three-day rising trend is detected, a **pattern penalty of +3** applies. Substituting into the formula of Section 3.3.4:

```
finalScore = clamp( (58 + (-4)) × 1.08 - 0 - 3 , 0, 100 )
           = clamp( 54 × 1.08 - 3 , 0, 100 )
           = clamp( 58.32 - 3 , 0, 100 )
           = clamp( 55.32 , 0, 100 )
           = 55    →   Level: Moderate
```

Because all thirteen drivers had usable input, aggregate coverage is 100 %, so the score is reported with **High** confidence. The interface would render this as a score of 55 in the "Moderate" band, with the largest contributors — sleep, screen time, and caffeine — surfaced as the top modifiable factor cards. This worked example illustrates the auditability claim of Section 3.3.7: the final number is never a verdict from a black box but the traceable result of an explicit calculation.

# Chapter 4: System Design and Model Design

This chapter describes the software architecture that realises the requirements of Chapter 3, the navigation model, and the persistent data model. The guiding architectural principle is a strict separation between a pure computational core and the device-facing layers that feed it, so that the scoring engine remains testable and the user interface remains a thin renderer of decomposed results.

## 4.1 Architectural Overview

Cadence follows a layered **Model–View–ViewModel (MVVM)** pattern augmented by a dedicated service layer and organised into feature modules. The application shell establishes a SwiftData model container and drives a top-level state machine from a splash screen through onboarding to the main tabbed interface. Below the shell, three collaborating layers — feature modules, a service layer of roughly thirty-five services, and the SwiftData model layer — interact with external integrations and a shared app group that bridges the main application and its three extensions. Figure 4.1 depicts this structure.

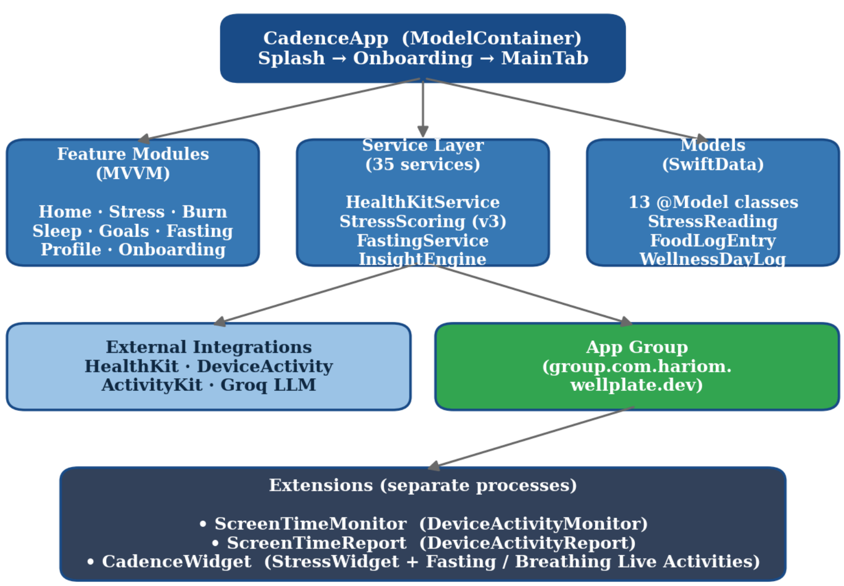

*Figure 4.1 — High-level system architecture, showing the MVVM feature modules, the service layer, the SwiftData models, the external integrations, and the three extension processes bridged through a shared app group.*

## 4.2 Design Patterns

- **MVVM with a service layer.** Each feature owns its views and view-models; all view-models are main-actor-isolated classes exposing observable published properties, eliminating a class of concurrency defects at compile time.
- **Service factory pattern.** Factories return either real or mock implementations of services such as the API client and the HealthKit service, selected by a build-time configuration flag, which enables deterministic testing and offline development.
- **Single source of truth for sheets.** Each feature defines one enumeration describing its modal sheets and drives a single sheet presenter, preventing the "stacked sheets" defect in which competing presentation modifiers race.
- **Pure scoring core.** The scoring engine is free of side effects and device dependencies; view-models gather inputs and feed them in, and the engine returns an array of per-driver results.

## 4.3 Navigation Model

The root view owns a high-level state machine. A splash screen is shown briefly; if the application is being launched for the first time, an onboarding flow is presented; thereafter, and on all subsequent launches, the main tabbed interface is shown. Tab selection is centralised in an observable selector so that deep links — for example, opening the Stress tab directly to the Fasting feature — function consistently across the application. Figure 4.2 illustrates this flow.

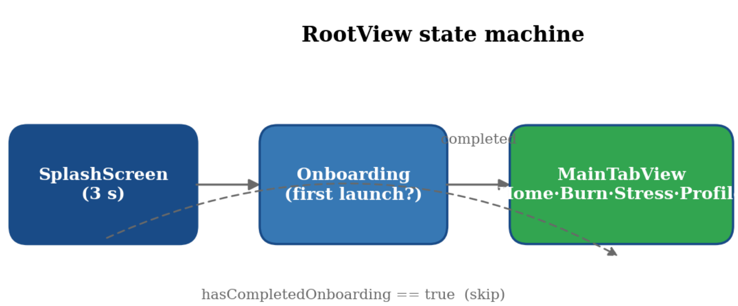

***Figure 4.2*** *— The RootView state machine governing the transition from splash to onboarding to the main tabbed interface.*

The main interface presents four tabs — Home, Burn, Stress, and Profile — using the modern iOS tab interface. The Stress tab is the flagship surface, hosting the decomposed daily score, the vitals panel, and the guided interventions.

## 4.4 Data Model

Persistence is implemented entirely with **SwiftData**, with thirteen model classes registered in the model container. These are complemented by approximately fourteen non-persisted value types — including the stress level enumeration, the per-driver result type, sleep and burn summaries, and the nutritional-information and data-transfer types — that carry computed or transient data. Table 4.1 summarises the persistent model classes and their responsibilities.

| Model class | Purpose |
|---|---|
| FoodLogEntry | A single meal: name, nutritional information, timestamp, meal type and context |
| FoodCache | Cached nutrition responses to avoid duplicate model calls |
| WellnessDayLog | Daily aggregate: water and coffee intake, notes, mood |
| UserGoals | Per-day-of-week targets, macros, and serialised home-layout configuration |
| StressReading | Intraday score samples with timestamp and confidence |
| InterventionSession | A breathing, relaxation, or journaling session record |
| FastingSchedule | The active fasting schedule (preset, custom hours, active flag) |
| FastingSession | A single fast: start, end, and broken-early flag |
| JournalEntry | Free text with timestamp and originating prompt |
| SymptomEntry | A symptom code with severity and timestamp |
| SupplementEntry | A supplement with dosage and timestamp |
| AdherenceLog | Whether a given goal was met on a given day |
| ManualDailyInput | Manually entered values when a framework source is unavailable |

***Table 4.1*** *— The thirteen persistent SwiftData model classes.*

## 4.5 Score-to-Level Mapping and Visual Design

The numeric score is mapped to a five-band qualitative level, each with a distinct visual accent of increasing intensity, so that the user perceives their state at a glance before examining the decomposition. Table 4.2 records this mapping. The design system uses a light grouped background, white cards with soft shadows and large corner radii, a consistent typographic ramp, blue as the primary accent, and green reserved for positive and completion states.

| Score range | Level | Visual accent |
|---|---|---|
| 0 – 20 | Excellent | Calm blue, low intensity |
| 21 – 40 | Good | Blue, moderate-low intensity |
| 41 – 60 | Moderate | Blue, medium intensity |
| 61 – 80 | High | Blue, high intensity |
| 81 – 100 | Very High | Solid blue, full intensity |

***Table 4.2*** *— Mapping of the numeric score to qualitative levels and visual accents.*

## 4.6 Concurrency and Error-Handling Design

All view-models are isolated to the main actor, so user-interface state is mutated only on the main thread; this eliminates an entire class of data races at compile time under Swift's concurrency model. Long-running work — HealthKit queries, screen-time aggregation, and network calls to the nutrition service — is dispatched to structured background tasks whose results are marshalled back to the main actor before publication. Errors are modelled explicitly rather than via untyped exceptions: the networking layer defines a typed error enumeration covering missing credentials, invalid responses, request timeouts, transport failures, and decoding failures, and each case maps to a specific, user-readable message. This explicit error surface is what allows the scoring engine to treat "missing input" as a first-class, gracefully-handled state rather than a failure.

## 4.7 Daily Update Sequence

A representative daily-score update proceeds as an ordered sequence. The home view-model requests the day's inputs from the service layer; the HealthKit service returns the available vitals, sleep, and activity aggregates; the screen-time values are read from the shared app group; manual logs and subjective entries are loaded from the SwiftData store. These inputs are assembled into the engine's input structure, with absent values flagged rather than defaulted. The pure scoring engine then computes the per-driver contributions, applies the four modifiers in order, clamps the result, and returns the decomposed result and confidence level. Finally, the view-model publishes the result, the interface renders the factor cards, and a `StressReading` sample is persisted so that the multi-day pattern detector has the history it needs on subsequent days. Because every step after input assembly is deterministic, the same sequence executed in a widget timeline or a unit test produces an identical score.

## Chapter 5: System Implementation

This chapter documents the realisation of the design in working software: the technology stack, the implementation of the scoring pipeline, the external integrations, the project structure, and the privacy posture. Figures and tables illustrate the concrete outcomes of the implementation.

### 5.1 Technology Stack

The application targets iOS 18 and above and is written in Swift with modern concurrency enabled by default. Table 5.1 summarises the technologies employed at each layer of the system.

| Layer | Technology | Notes |
|---|---|---|
| Language | Swift 5 | Modern concurrency; main-actor isolation by default |
| User interface | SwiftUI | Modern tab interface; a unified typographic and shadow system |
| Charts | Swift Charts | Stress trends, sleep stages, and activity bars |
| Persistence | SwiftData | Thirteen model classes; no Core Data |
| Health data | HealthKit | Steps, heart rate, HRV, sleep, blood pressure, energy |
| Screen time | DeviceActivity / FamilyControls | Extension writes thresholds to a shared app group |
| Live Activities | ActivityKit | Fasting and breathing Dynamic Island surfaces |
| Nutrition NLP | Groq Cloud (llama-3.3-70b) | Fast natural-language meal parsing |
| Concurrency | async/await, structured tasks | Main-actor view-models; background helpers |
| Distribution | App + 3 extensions | Screen-time monitor, screen-time report, widget |

***Table 5.1** — The technology stack by architectural layer.*

### 5.2 Implementation of the Scoring Pipeline

The deterministic pipeline of Section 3.3 is implemented as a single pure function. View-models assemble the day's inputs from the service layer, the engine computes each driver's contribution, sums them, applies the four modifiers in the prescribed order, clamps the result, and returns the decomposed array of per-driver results together with the confidence level. Figure 5.1 visualises this computation as a pipeline.

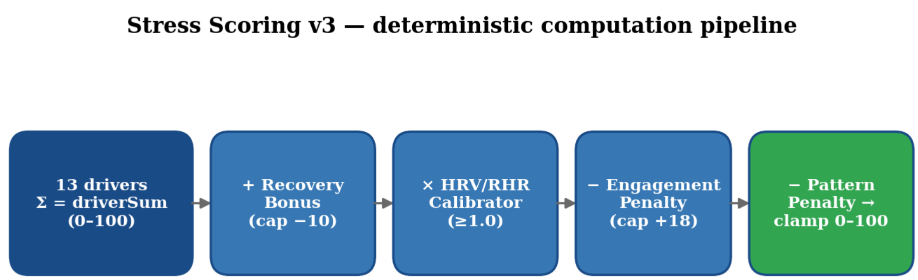


***Figure 5.1*** *— The deterministic computation pipeline: the thirteen-driver sum is adjusted by the recovery bonus, scaled by the physiological calibrator, reduced by the engagement and pattern penalties, and clamped to the valid range.*

Because the engine is stateless, the same pipeline executes identically whether invoked from the live application, a widget refresh, or a unit test, which is the foundation of both the auditability claim of Section 3.3.7 and the testing strategy of Chapter 6. The engine's surface is a single pure entry point that takes a fully-assembled input value and returns a decomposed result, as illustrated below.

```
// Pure, stateless scoring entry point (illustrative signature)
struct StressScoringEngine {
    static func score(_ input: StressInput) -> StressResult {
        let drivers = computeDrivers(input)      // 13 contributions
        let base    = drivers.reduce(0) { $0 + $1.points }
        let bonus   = recoveryBonus(input)         // capped at -10
        let cal     = calibrator(input)            // >= 1.0
        let engage  = engagementPenalty(input)     // capped at +18
        let pattern = patternPenalty(input)        // capped at +12
        let raw     = (Double(base + bonus) * cal) - engage - pattern
        return StressResult(score: clamp(raw, 0, 100),
                            drivers: drivers,
                            confidence: confidence(for: drivers))
    }
}
```

## 5.3 External Integrations

### 5.3.1 HealthKit

The HealthKit service reads steps, active energy, heart rate, resting heart rate, HRV expressed in milliseconds, systolic and diastolic blood pressure, respiratory rate, sleep analysis, and daylight exposure. Background delivery is enabled by entitlement so the application can refresh without being foregrounded. Vitals are read as daily averages rather than sums where averaging is the physiologically meaningful aggregation. All HealthKit data remains on the device.

### 5.3.2 DeviceActivity (Screen Time)

Screen-time measurement uses the DeviceActivity framework, which requires Apple's restricted Family Controls entitlement. A monitor extension registers hourly thresholds and, when a threshold fires, writes total and evening usage to the shared app group. A separate report extension renders a category breakdown inside the application. Evening usage, defined as the hours from early evening to midnight, is weighted more heavily in the scoring engine because of its documented effect on melatonin and sleep [13], [14].

### 5.3.3 ActivityKit Live Activities

Two Live Activities are implemented. The fasting Live Activity presents a progress ring, elapsed and remaining time, and the eat-window endpoint on the Lock Screen and in the Dynamic Island, including a compact watch variant. The breathing Live Activity is active during a guided session. Both update on state transitions rather than on a per-second timer, conserving the system's Live Activity update budget.

### 5.3.4 Large-Language-Model Nutrition Parsing

Natural-language meal descriptions are parsed by a hosted large-language model from the Llama family [30], accessed through the Groq inference service. A system prompt constrains the model to return structured nutritional information as machine-readable output; a user prompt carries only the meal description the user typed. Responses are cached to avoid duplicate calls, and errors are surfaced through a typed error model covering missing credentials, invalid responses, timeouts, network failures, and decoding failures. In debug builds a mock client returns bundled sample data, decoupling development from the network.

## 5.4 Project Structure

The codebase is organised by responsibility, with an application shell, a core services directory, feature modules, models, networking, shared design assets, and the widget and extension targets. Files are auto-discovered by the build system's file-system-synchronised grouping, which eliminates project-file merge conflicts. The principal directory layout is shown below.

```
Cadence/
  App/            Entry point + RootView state machine
  Core/
    AppConfig     Build toggles, model selection, timeouts
    Services/     ~35 services (HealthKit, StressScoring, ...)
  Features/       Home, Stress, Burn, Sleep, Goals, Fasting,
                  Onboarding, FoodScanner, History, Symptoms,
                  Supplements, Progress, Profile, Tab
  Models/         SwiftData @Model classes + value types
  Networking/     Real + Mock API clients (factory-selected)
  Shared/         Colour tokens, components, extensions
  Widgets/        Live Activity attribute types
Extensions:       ScreenTimeMonitor, ScreenTimeReport, Widget
```

## 5.5 Entitlements and Privacy Posture

The application requests only the entitlements strictly necessary for its features, listed in Table 5.2. The privacy posture is deliberately conservative and aligns with the recommendations of the mHealth privacy literature [29]: all HealthKit data remains on the device; the only outbound network call carries the meal text the user explicitly entered, containing no protected health identifiers or device identifiers; and credentials are stored outside source control and loaded at runtime.

| Capability | Purpose |
|---|---|
| HealthKit | Read vitals, sleep, and activity data |
| HealthKit background delivery | Refresh data when the app is not foregrounded |
| Family Controls | DeviceActivity screen-time monitoring (restricted entitlement) |
| App Groups | Share data between the main app and its extensions |

***Table 5.2*** *— Requested entitlements and their purposes.*

## 5.6 Development Workflow

Feature work is gated through a structured multi-stage workflow — from brainstorming and strategy through planning, auditing, resolution, and checklisting to implementation and fixing — with each stage producing a dated artifact, enforcing a documented paper-trail for every non-trivial change. Build verification is performed across the four schemes of the main application and its three extensions, ensuring that a change which compiles for the app also compiles for every dependent extension.

## 5.7 Food Logging and the Insight Engine

Two feature subsystems illustrate how the service layer feeds the scoring engine. The **food-logging** subsystem lets the user describe a meal in natural language; the description is sent to the large-language-model service, which returns structured nutritional information that is persisted as a `FoodLogEntry` and cached in a `FoodCache` record to avoid duplicate calls for repeated meals. The parsed entries inform the Diet, Caffeine, Eating-Triggers, and Meal-Timing drivers — for example, a late, high-sugar meal logged after a stressful afternoon contributes to both the meal-timing and eating-trigger signals, consistent with the stress-eating literature [19].

The **insight engine** consumes the daily wellness record and the decomposed stress result to generate plain-language observations — for instance, noting that evening screen time has exceeded its usual range on several recent days, or that hydration has been consistently low. These insights are not generic tips but are derived from the user's own logged drivers, operationalising the just-in-time adaptive intervention principle [23] by surfacing a specific, modifiable factor at the moment it begins to trend adversely. Crucially, because the underlying score is decomposable, every insight can name the exact driver it is responding to.

## Chapter 6: Testing

Testing focuses on the component whose correctness matters most and is most amenable to rigorous verification: the deterministic scoring core. Because that engine is pure and stateless, it can be exercised exhaustively in isolation, which is the principal reason the architecture was designed to keep it free of device dependencies.

### 6.1 Testing Strategy

The strategy combines three complementary layers. **Unit testing** verifies the deterministic core and the networking orchestration against pinned, hand-computed scenarios. **Mock-based testing** exercises the data-parsing paths using bundled sample responses, removing any dependence on live network services. **Build verification** confirms that the entire system — the main application and all three extensions — compiles cleanly, catching integration regressions that unit tests alone cannot.

### 6.2 Unit Test Coverage

The unit-test suite targets the scoring engine and the nutrition pipeline. Table 6.1 summarises the principal test classes and the behaviour each verifies.

| Test class | Verifies |
|---|---|
| Stress-scoring tests | Pinned scenarios for each driver and each modifier, and the final clamped formula |
| Nutrition-service tests | Orchestration across real and mock providers |
| Nutrition-provider tests | Request construction and response decoding |
| Mock-provider tests | Correct parsing of bundled sample data |
| Meal-transcription tests | The speech-to-text meal-entry path |

***Table 6.1*** *— Principal unit-test classes and the behaviour each verifies.*

### 6.3 Determinism as a Testability Property

The determinism established in Section 3.3.7 is not merely a transparency feature; it is the property that makes the scoring engine straightforwardly testable. Each pinned scenario specifies a complete table of inputs and the exact expected score, and because the engine is free of randomness and hidden state, the assertion is stable across runs and machines. A learned model would require statistical tolerance bands and a maintained test dataset; the deterministic engine requires neither.

## 6.4 Test Scenarios

Representative scenarios include: a fully-covered "good day" producing a low score with high confidence; a sleep-deprived, high-caffeine, high-evening-screen-time day producing a markedly higher score; a sparsely-logged day producing a medium or low confidence level with an appropriate engagement penalty; a day with completed interventions confirming that the recovery bonus is applied and capped; and a multi-day rising sequence confirming that the pattern penalty engages. Each scenario asserts both the final score and the individual driver contributions, ensuring that the decomposition the user sees is itself tested.

## 6.5 Limitations of the Current Test Suite

Automated user-interface testing is not yet wired into the shared schemes; interface behaviour is presently verified manually. Integration with the live HealthKit and DeviceActivity frameworks is verified by build and manual on-device testing rather than by automated tests, because these frameworks require entitlements and real device data that are impractical to simulate in a continuous-integration environment. Extending coverage to automated interface and integration tests is identified as future work in Chapter 8.

## Chapter 7: Results and Discussion

This chapter reports the outcomes of the project against its stated objectives and discusses them in the light of the literature. Because Cadence is a deterministic decision-support tool rather than a learned classifier, its results are assessed not by a single accuracy figure but by the degree to which it satisfies the functional and non-functional requirements of Chapter 3 and improves on the transparency and intervention limitations of existing systems.

### 7.1 Functional Outcomes

All six objectives stated in Section 1.4 were met. The application computes an explainable 0–100 daily stress score from thirteen heterogeneous drivers; presents each driver as a modifiable factor card; persists a complete daily record locally with no mandatory backend; provides prospective interventions including guided breathing, progressive muscle relaxation, fasting protocols, and journaling; confines all HealthKit data to the device; and degrades gracefully to a neutral physiological multiplier when wearable data is absent. The system comprises a main application and three extensions that build cleanly across all schemes.

### 7.2 The Explainability Outcome

The central result is qualitative but consequential: every score is fully decomposable. When a user asks why their score rose, the interface can attribute the change to specific drivers — for example, a fourteen-point increase decomposed as eight points from reduced sleep, four from elevated evening screen time, and two from a missed exercise target. This stands in direct contrast to commercial scores, whose proprietary formulas cannot offer such an attribution. This outcome operationalises the interpretability principles advocated by Markus et al. [20] and Rudin [21], and demonstrates that a transparent model can deliver genuinely actionable feedback rather than an opaque verdict.

### 7.3 Illustrative Stress Trajectory

Figure 7.1 presents an illustrative weekly trajectory that demonstrates the system's explanatory capability. The mid-week peak is attributable to a combination of poor sleep and high evening screen time, while the weekend decline reflects completed exercise and breathing interventions registering through the recovery bonus. The figure is illustrative of the decomposition mechanism rather than a clinical measurement.

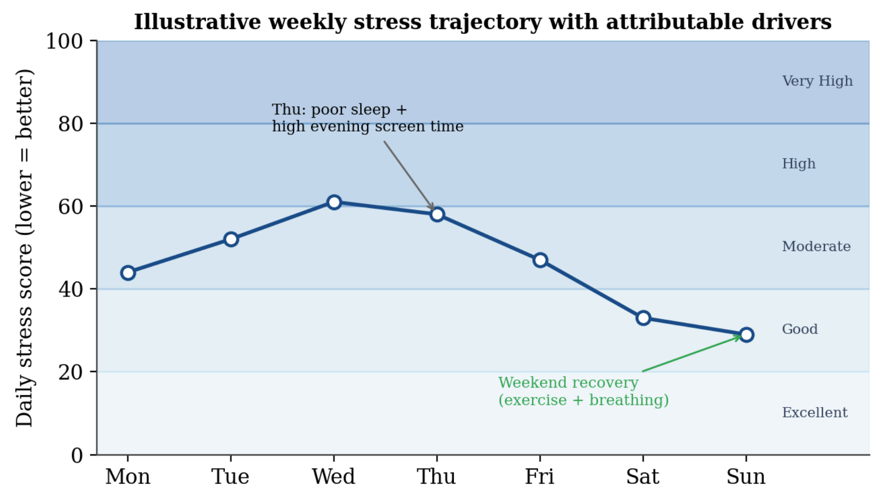


***Figure 7.1** — An illustrative weekly stress trajectory in which each inflection is attributable to specific drivers, demonstrating the explanatory value of the decomposed score.*

## 7.4 Comparison with Commercial Systems

Table 7.1 compares Cadence with the dominant commercial recovery-score systems along the dimensions most relevant to this project's thesis. The comparison is drawn from the peer-reviewed device-validation literature [26], [27], [28] and from the publicly documented behaviour of each system; the underlying scoring formulas of the commercial systems are proprietary and unpublished, which is itself the salient point of comparison.

| Dimension | Commercial HRV scores | Cadence |
|---|---|---|
| Primary signal | Overnight HRV / RHR | 13 multimodal drivers |
| Behavioural drivers | Largely absent | Central to the model |
| Score transparency | Proprietary / opaque | Fully decomposable |
| Intervention timing | Retrospective | Prospective |
| On-device privacy | Cloud-dependent (varies) | On-device by default |
| Dedicated hardware | Required | Optional (degrades gracefully) |

***Table 7.1** — Comparison of Cadence with commercial HRV-centric recovery-score systems.*

## 7.5 Discussion: Multimodal Fusion versus Single-Signal Scoring

The design decision to fuse thirteen drivers rather than rely on HRV alone is supported by the multimodal-detection literature. The WESAD study reported approximately 93 % binary stress-detection accuracy from fused chest-sensor signals [7], and free-living wrist-plus-context monitoring established the practical value of combining physiology with behaviour [8]. Figure 7.2 situates Cadence's multimodal philosophy within this evidence; the multimodal bar is anchored to the WESAD result, while the single-signal and behaviour-only bars are illustrative of the qualitative gap rather than measured values for Cadence itself.

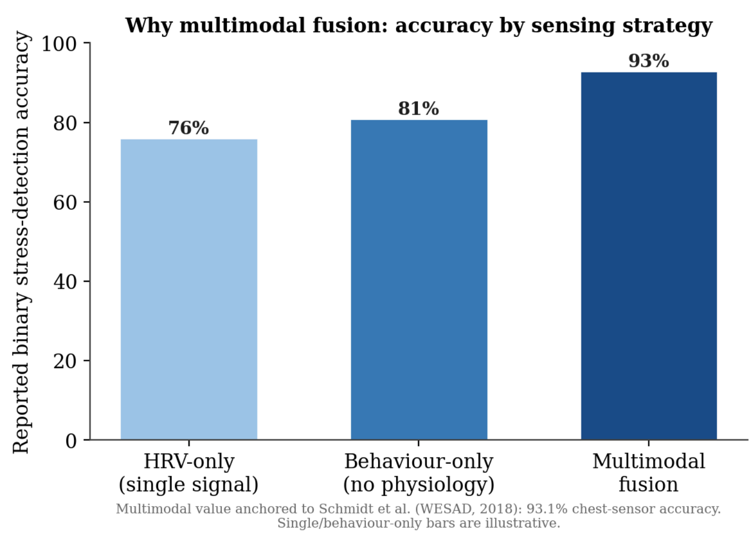


***Figure 7.2*** *— The rationale for multimodal fusion: reported stress-detection accuracy is higher for fused signals than for single-signal approaches, with the multimodal value anchored to the WESAD benchmark.*

The practical advantage of the multimodal approach is not only higher detection accuracy but earlier intervention. Because behavioural drivers such as evening screen time are observable in real time, Cadence can flag a developing stressor before it propagates into disrupted sleep and depressed HRV — closing the prospective-intervention gap identified in Section 1.2 and consistent with the just-in-time adaptive intervention paradigm [23].

## 7.6 Discussion: Determinism, Privacy, and Trust

Two non-functional outcomes reinforce the system's trustworthiness. The deterministic core means the model can be audited from a single table of inputs, satisfying the standard urged for high-stakes health algorithms [21], [22]. The on-device privacy posture aligns with the recommendations of the mHealth privacy literature [29] and removes a barrier to adoption for privacy-sensitive users. Together, transparency and privacy distinguish Cadence from systems that are both opaque and cloud-dependent.

## 7.7 Threats to Validity and Limitations

Several limitations temper these results. First, the driver weights, though grounded in the literature, are fixed expert priors rather than personalised values; an individual's true sensitivity to, say, caffeine may differ from the population average, and the caffeine literature itself notes dose- and tolerance-dependent effects [12]. Second, the score has not yet been validated against a gold-standard stress measure such as salivary cortisol or a validated questionnaire administered longitudinally; such validation is essential future work. Third, some inputs depend on user self-report, which is subject to recall and compliance bias. Fourth, the evidence for certain modulators — notably time-restricted eating — remains mixed in the wider literature [18], so their modest weights are deliberately conservative. These limitations define the validation agenda set out in the next chapter.

## 7.8 Realised User-Facing Surfaces

Beyond the algorithm, the project delivered a complete set of user-facing surfaces that make the score actionable. The **Stress view** presents the daily score within its qualitative band and expands into the ranked factor cards, each of which doubles as an entry point for the corresponding intervention. The **home-screen widget** renders the current score and band without opening the application, and the **Live Activities** surface an active fast or a guided breathing session on the Lock Screen and in the Dynamic Island. The **food scanner** provides the natural-language meal-logging entry point, and the **fasting tracker** offers multiple presets with a progress ring. Together these surfaces close the loop from measurement to action: a rising driver is not merely reported but is paired, at the point of display, with the specific intervention that addresses it. This realises the prospective-intervention objective of Section 1.4 and distinguishes Cadence from systems that report a score without offering a lever to change it.

# Chapter 8: Conclusion and Future Scope

## 8.1 Conclusion

This project set out to determine whether a consumer mobile application could compute a single, trustworthy, and explainable measure of daily psychological stress from heterogeneous signals while preserving privacy and remaining useful when wearable data is incomplete. The answer, demonstrated by the design and implementation of Cadence, is affirmative. By fusing thirteen behavioural, physiological, and lifestyle drivers in a transparent, tier-weighted, deterministic engine, the application produces a score that is not only computed but *explained*, attributing every change to specific, modifiable inputs.

In doing so the project makes three contributions. It demonstrates a multimodal alternative to the prevailing single-signal commercial paradigm, grounded in the clinical literature on each driver. It shows that a deterministic, fully auditable model can deliver actionable feedback, in keeping with the principles of interpretable health AI. And it realises a privacy-preserving, on-device architecture that integrates several privileged Apple frameworks without surrendering sensitive health data to the cloud. The principal limitation — that the score awaits longitudinal validation against a physiological gold standard — defines the immediate next step rather than undermining the contribution, since the deterministic design makes such validation straightforward to conduct and interpret.

## 8.2 Future Scope

Several avenues extend this work:

- **Apple Watch companion application** to surface intraday stress and feed completed relaxation sessions back into the recovery bonus.
- **iCloud synchronisation** of the SwiftData store, which is presently local-only, to support multi-device use while preserving end-to-end privacy.
- **HealthKit write-back** of guided breathing and relaxation sessions as mindfulness workouts, contributing to the user's wider health record.

- **On-device CoreML calibration** of personal driver weights learned from the user's own logged history, retaining transparency by constraining the learned component to interpretable per-driver scaling factors.
- **Longitudinal validation** of the score against salivary cortisol and validated self-report instruments such as the Perceived Stress Scale [3], to establish criterion validity.
- **Optional, opt-in, privacy-preserving aggregation** to support population-level studies of driver impact, using privacy-preserving techniques drawn from the federated mHealth literature [29].

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